TCES 330

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Part 1:

Translate the following pseudocode into a sequence of legal instructions; represent machine code in hex.
RF[0A] = D[1B] - D[2A] + D[3C] - D[7E];

D[6A] = RF[0A];

HALT

asm:	%machine code	#HEX
LDR 1B 0	0010 00011011 0000	21B0
LDR 2A 1	0010 00101010 0001	22A1
LDR 3C 2	0010 00111100 0010	23C2
LDR 7E 3	0010 01111110 0011	27E3
SUB 0 1 0	0100 0000 0001 0000	4010
ADD 0 2 0	0011 0000 0010 0000	3020
SUB 0 3 0	0100 0000 0011 0000	4030
STR 0A 6A	0001 1010 0110 1010	1A6A
HLT	0101 0000 0000 0000	5000

Part 2:

The ALU module was tested by linking it to the DE-2 board according to the description below. The Verilog module was tested for each function, and 256 randomized instructions and data inputs.

ALU to DE-2 hardware mapping, taken from SV module header

The first 16 switches set either aluA (shown on hex7-4) or aluB (shown on hex3-0) registers, based on the state of switch 16. The registers then feed into the inputs of the ALU, which has its operation run on by CLOCK_50. The alu then is operated by positive clock edge to output to the 16 red LEDs below the switches. KEY 3-0 (on LEDG 3-0) determine the ALUop.